

Exp No:12 Experiment to detect outliers in a given data set.

Description: Understand the procedure to identify the outliers in a given dataset

```
#sample calculation for low range(lr) , upper range (ur),percentile
import numpy as np
array=np.random.randint(1,100,16) # randomly generate 16 numbers between 1 to 100
array
```

```
→ array([27, 50, 44,  6, 58, 61, 23, 86, 67, 20, 75,  7, 79, 61, 90, 54])
```

```
array.mean()
```

```
→ 50.5
```

```
np.percentile(array,25)
```

```
→ 26.0
```

```
np.percentile(array,50)
```

```
→ 56.0
```

```
np.percentile(array,75)
```

```
→ 69.0
```

```
np.percentile(array,100)
```

```
→ 90.0
```

```
#outliers detection
def outDetection(array):
    sorted(array)
    Q1,Q3=np.percentile(array,[25,75])
    IQR=Q3-Q1
    lr=Q1-(1.5*IQR)
    ur=Q3+(1.5*IQR)
    return lr,ur
```

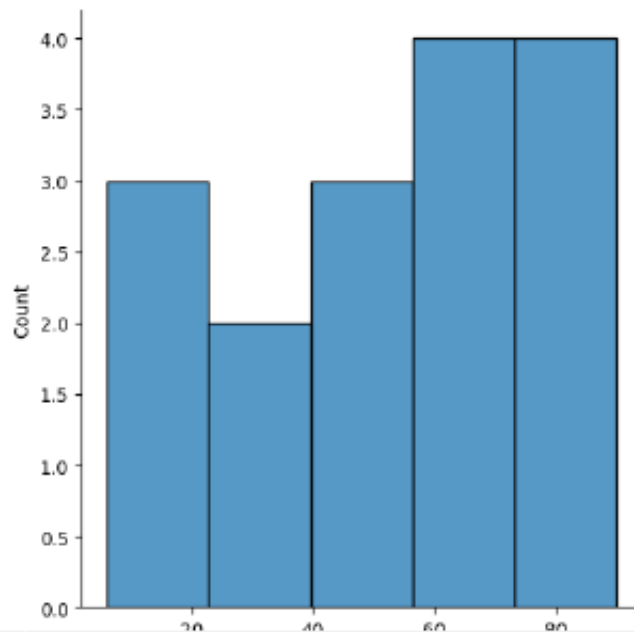
```
lr,ur=outDetection(array)
```

```
lr,ur
```

```
→ (-38.5, 133.5)
```

```
import seaborn as sns
%matplotlib inline
sns.displot(array)
```

```
→ <seaborn.axisgrid.FacetGrid at 0x78f3291c2710>
```



```
sns.distplot(array)
```

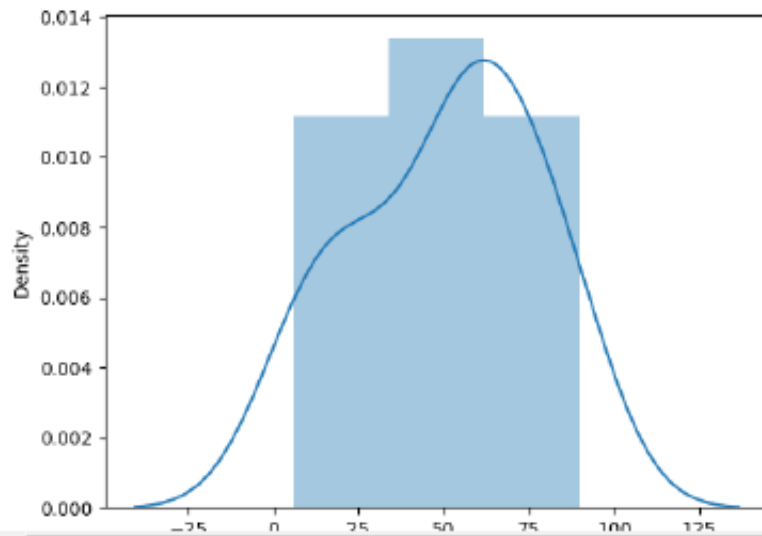
 <ipython-input-19-d72101983c40>:1: UserWarning:

'distplot' is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either 'displot' (a figure-level function with similar flexibility) or 'histplot' (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see <https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751>

```
sns.distplot(array)
<Axes: ylabel='Density'>
```

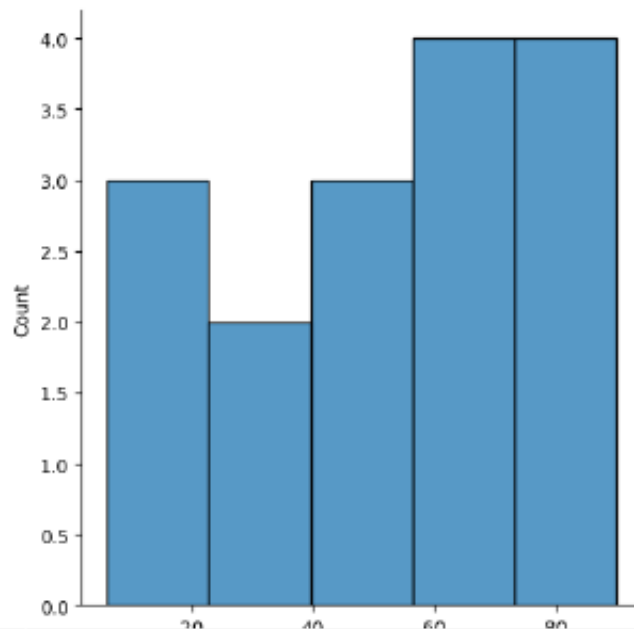


```
new_array=array[(array>1r) & (array<ur)]
new_array
```

 array([27, 50, 44, 6, 58, 61, 23, 86, 67, 20, 75, 7, 79, 61, 90, 54])

```
sns.displot(new_array)
```

 <seaborn.axisgrid.FacetGrid at 0x78f2e09bb580>



```
lr1,ur1=outDetection(new_array)
lr1,ur1
```

 (-38.5, 133.5)

```
final_array=new_array[(new_array>lr1) & (new_array<ur1)]
final_array
```

 array([27, 50, 44, 6, 58, 61, 23, 86, 67, 20, 75, 7, 79, 61, 90, 54])

```
sns.distplot(final_array)
```

 <ipython-input-18-7ba96ada5b76>:1: UserWarning:

'distplot' is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either 'displot' (a figure-level function with similar flexibility) or 'histplot' (an axes-level function for histograms).

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```
sns.distplot(final_array)  
<Axes: ylabel='Density'>
```

